Validation Proportion Matters: Generalization Stability of YOLO-Based Wafer Defect Detection with Limited Data

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***Abstract*— Automated wafer surface inspection increasingly relies on deep learning, yet limited annotated datasets challenge generalization stability. This study investigates how validation proportion affects YOLO-based defect detection using a 300-image microscopy dataset with three defect classes: small-area undercut (sUdc), large-area undercut (lUdc), and particles (Ptc). YOLOv8m and YOLOv11m were evaluated under 70/15/15, 70/20/10, and 80/10/10 train/validation/test splits. Under identical training conditions, YOLOv11m with a 70/20/10 split achieved the best performance (mAP50 = 0.819, Recall = 0.810). Training with 80% data did not improve performance and decreased validation stability. Results demonstrate that, in limited semiconductor datasets, maintaining adequate validation sampling is more critical than maximizing training size.**

Keywords— YOLO, AdamW, Semiconductor, Defect Detection

I. Introduction

Surface defect inspection plays a central role in semiconductor process control. Defects such as undercut formation and particulate contamination can degrade structural integrity, distort lithographic patterns, and negatively affect device yield [1]–[3]. Traditional rule-based inspection systems are sensitive to illumination variation and surface texture complexity, limiting robustness in advanced fabrication environments.

Deep learning–based object detection frameworks, particularly YOLO (You Only Look Once) [4], have demonstrated strong real-time detection capability across industrial applications. Recent architectural improvements have enhanced feature fusion, small-object localization, and computational efficiency [5]. However, semiconductor inspection datasets are typically small due to acquisition constraints and manual labeling costs.

Under limited-data conditions, model generalization is highly sensitive to dataset partitioning. While increasing the training proportion is commonly assumed to improve performance, insufficient validation sampling may lead to unstable convergence monitoring and unreliable model selection. Despite its practical importance, the influence of validation proportion on generalization stability remains largely unexplored in semiconductor defect detection.

This study presents a controlled experimental evaluation of YOLOv8m and YOLOv11m under three dataset partition configurations. The primary contribution is demonstrating that, for small semiconductor datasets, maintaining a sufficiently large validation set (20%) yields more stable and reliable generalization than increasing the training proportion to 80%.

II. Experimental Setup

A. Dataset Acquisition

The dataset consists of 300 high-resolution wafer microscopy images captured at 5× magnification under cleanroom illumination. Three annotated defect classes were included:

- Small-area undercut (sUdc)
- Large-area undercut (lUdc)
- Particulate contamination (Ptc)

Undercut defects result from isotropic etching effects, while particles originate from contamination sources [1]–[3].

Three dataset partition strategies were evaluated:

- 70% training / 15% validation / 15% testing
- 70% training / 20% validation / 10% testing
- 80% training / 10% validation / 10% testing

These configurations were selected to analyze the trade-off between training capacity and validation reliability under limited data conditions.

B. Model Architecture and Training

Two medium-scale YOLO architectures were evaluated:

- YOLOv8m
- YOLOv11m

YOLO formulates object detection as a single-stage regression problem [4]. YOLOv11m incorporates architectural refinements to improve detection sensitivity and parameter efficiency [5].

All experiments were conducted using identical hyperparameters:

- Optimizer: AdamW [6]
- Maximum epochs: 200
- Early stopping: 50-epoch patience

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- Initial learning rate: 0.0003
- Weight decay: 0.0005
- Batch size: 12
- Input resolution: 640×640

To address the specific requirements of semiconductor defect geometries, both YOLOv8m and YOLOv11m were customized using a transfer learning approach. While both models utilize a decoupled head design, the standard 80-class classification layer was replaced with a specialized 3-class linear layer (sUdc, lUdc, Ptc). Pre-trained weights from the COCO dataset were utilized as a backbone for feature extraction. Hyperparameters were kept at default values to ensure a consistent baseline, with the exception of the 50-epoch early stopping patience, which was implemented to prevent overfitting in a limited-data environment.

Data augmentation (training only) included brightness variation, grayscale perturbation, Gaussian blur, and additive noise to improve robustness under limited data [7].

Experiments were conducted using the Ultralytics YOLO framework [8] in a GPU-accelerated environment (NVIDIA RTX 4060 with cuDNN support [9]). Dataset management and annotation were performed using Roboflow [10], and training metrics were tracked via Weights & Biases for experiment monitoring and reproducibility [11].

Performance was evaluated using Precision, Recall, mAP50, and mAP50–95 following the Common Objects in Context (COCO) evaluation protocol [12], [13].

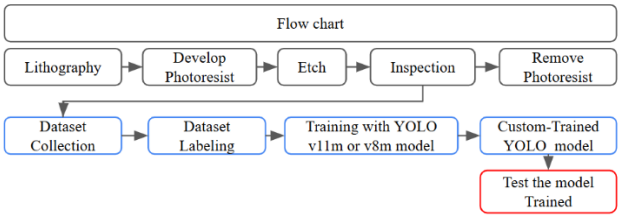


Fig. 1. Photolithography/Etch and Defect Training Flow Chart.

III. METHODOLOGY

A. Training Procedure

The overall training workflow is illustrated in Fig. 1. Both YOLOv8m and YOLOv11m were trained under identical experimental conditions to ensure a controlled comparative study. The same dataset, image resolution, batch size, optimization parameters, and evaluation metrics were maintained across all experiments. Only the model architecture and data split configuration were varied

B. Data Augmentation

To enhance robustness and generalization capability, data splitting was performed on the original 300 images prior to any augmentation. This prevents data leakage, ensuring that the validation and test sets remain strictly composed of original, non-augmented images. For the training partition, a factor of 3 times augmentation was applied.

Preprocessing included auto-orientation and resizing all images to a fixed resolution of 640 × 640 pixels using bilinear interpolation; for non-square aspect ratios, empty space was filled with white padding. A global histogram equalization was applied to the training set to highlight the boundaries of undercut defects. The specific augmentations applied to 15% of the training images included:

- The following augmentation techniques were applied:
- Grayscale conversion (15% probability)
- Brightness adjustment (±15%)
- Gaussian blur (kernel equivalent of 1.5 pixels)
- Additive noise (0.1% intensity)

These augmentations simulate illumination variation, microscope focus variability, and sensor noise, thereby improving robustness in real-world inspection scenarios. Augmentation strategies were selected based on established practices for improving deep learning generalization under limited data conditions.

C. Defect Class Definitions

To improve visualization clarity (Fig 3.), abbreviated labels were used during detection output visualization.

1) Small-area undercut (sUdc)

Small-area undercut refers to localized lateral etching beneath the photoresist (PR) layer. Undercut formation is commonly associated with isotropic etching processes, where material removal occurs uniformly in vertical and lateral directions, resulting in unintended sidewall recession beneath the PR mask [1], [2].

2) Large-area undercut (lUdc)

Large-area undercut describes the coalescence of multiple small undercut regions into a continuous, extended lateral etch region. This may occur due to prolonged etch duration, excessive chemical activity, or inadequate mask adhesion [1], [3]. In severe cases, a large-area undercut may spatially overlap with or fully encompass small-area undercut regions.

3) Particles (Ptc)

Particles represent foreign particulate contamination on the wafer surface. Such contamination may originate from airborne debris, process residues, equipment wear, or handling artifacts. Particles can interfere with photolithography, etching uniformity, and thin-film deposition processes, potentially leading to pattern distortion or device failure [1], [3]. Accurate detection of particulate defects is therefore critical for yield control and process reliability.

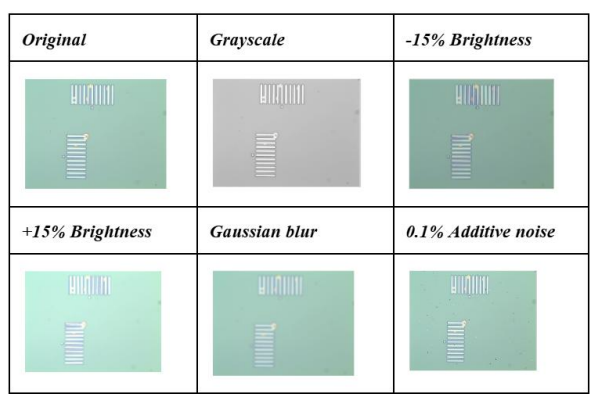


Fig. 2. Data Augmentation for Grayscale, Brightness, Blurring, and noise.

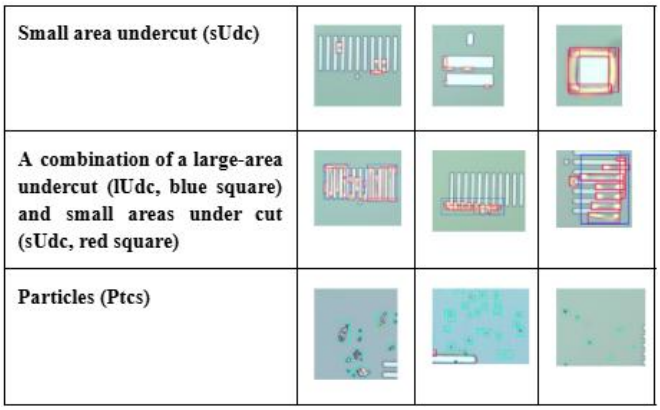


Fig. 3. Data labeling.

TABLE I. CLASS DISTRIBUTION

Class	Total Instances	Common Density	Peak Density
Ptc	~1200	2-25 instances	Up to 29 instances
sUdc	~2100	2–10 instances	Up to 37 instances
lUdc	~150	1 instance	Up to 3 instances

The rationale for prioritizing a 20% validation split is based on the instance-level distribution of defect classes. While a 10% split (30 images) might seem standard, it provides a statistically thin supervisory signal for rarer defects like lUdc, which appears in only a fraction of the total images. By allocating 20% (60 images) to validation, the model is exposed to a more representative sample of approximately 600+ total defect instances during the evaluation phase. This prevents the 'noisy' convergence and validation loss spikes observed in smaller partitions, ensuring that early stopping is triggered by true generalization rather than a localized artifact of a limited sample set.

IV. RESULTS AND TRAINING ANALYSIS

Two medium-scale object detection architectures were evaluated: YOLOv8m and YOLOv11m.

A. Learning Rate Scheduling and Optimization Behavior

The learning rate of parameter group 0 (lr/pg0) corresponds to the backbone parameter group in the YOLO optimization pipeline. In Ultralytics implementations, parameters are divided into groups (e.g., backbone, detection head, bias terms), each of which may be assigned different optimization configurations [11]. A progressive decay scheduling strategy was adopted to stabilize convergence and reduce oscillatory updates. Proper learning rate scheduling is critical for stable deep neural network optimization [15].

Across all configurations, final lr/pg0 values ranged between 1.0×10^{-4} and 1.9×10^{-4}, depending on architecture and dataset partition. Under the 70/20/10 split, YOLOv11m converged at epoch 93, whereas YOLOv8m converged earlier at epoch 28. The extended convergence period for YOLOv11m suggests that the deeper architecture benefited from longer training to fully exploit its hierarchical feature extraction capability. The AdamW optimizer [6] was employed to decouple weight decay from gradient updates, improving regularization stability and mitigating overfitting. This optimization strategy contributed to smooth convergence across all experimental settings.

A comparative analysis of loss metrics (Table 7) reveals that the 70/20/10 split provided the most stable optimization environment. For YOLOv11m, the total validation loss (*Box* + *Cls* + *Dfl*) was lower and more consistent in the 70/20/10 configuration compared to the 80/10/10 split, where the val/cls_loss spiked to 1.002. This indicates that when the validation set is reduced to only 10% (30 images), the supervisory signal for early stopping becomes noisy, leading to sub-optimal weight convergence. Conversely, the 20% validation buffer allowed YOLOv11m to continue refining features until epoch 93, nearly triple the training duration of the 70/15/15 split (35 epochs), resulting in the highest recorded mAP50 of 0.819.

TABLE II. OVERALL PERFORMANCE ACROSS MODELS AND DATA SPLITS

Model	Split	mAP50	mAP50–95	Precision	Recall
YOLOv8m	70/15/15	0.801	0.437	0.772	0.765
YOLOv8m	70/20/10	0.805	0.464	0.791	0.784
YOLOv8m	80/10/10	0.782	0.456	0.742	0.731
YOLOv11m	70/15/15	0.804	0.452	0.785	0.795
YOLOv11m	70/20/10	0.819	0.46	0.781	0.81
YOLOv11m	80/10/10	0.792	0.447	0.755	0.772

TABLE III. FINAL IMAGE DISTRIBUTION POST-AUGMENTATION (NO DATA LEAKAGE)

Split Strategy	Training (3 x Aug)	Validation (Original)	Test (Original)	Total Images
70/15/15	630	45	45	720
70/20/10	630	60	30	720
80/10/10	720	30	30	780

TABLE IV. OVERALL DETECTION PERFORMANCE ACROSS MODELS AND DATA SPLITS

Class	mAP50	mAP50–95
lUdc	0.956	0.72
sUdc	0.91	0.46
Ptc	0.591	0.199

TABLE V. COMPARATIVE EVALUATION OF 70/20/10 VS. 80/10/10 PARTITIONING ON YOLOV11M METRICS

Model	Split	mAP50	Recall	Val Cls Loss
YOLOv11m	70/20/10	0.819	0.810	0.960
YOLOv11m	80/10/10	0.791	0.773	1.002

B. Detection Performance Evaluation

Detection performance was evaluated using precision, recall, mean Average Precision at IoU = 0.5 (mAP50), and mean Average Precision across IoU thresholds from 0.5 to 0.95 (mAP50–95). Precision–Recall analysis is particularly appropriate for imbalanced detection tasks, as it better reflects classifier behavior across varying confidence thresholds than ROC-based evaluation [18]. The mAP50–95 metric follows the COCO evaluation protocol [12] and provides a stricter localization assessment than fixed-IoU benchmarks such as PASCAL VOC [14]. Definitions and interpretations follow standardized methodologies summarized in [13].

Among all configurations, YOLOv11m trained under the 70/20/10 split achieved the strongest overall performance:

- mAP50 = 0.819
- mAP50–95 = 0.460
- Precision = 0.781
- Recall = 0.810

Compared to YOLOv8m under the same configuration (mAP50 = 0.805; mAP50–95 = 0.464; Recall = 0.784), YOLOv11m demonstrated superior recall and improved detection balance. Higher recall is particularly important in semiconductor defect inspection, where missed detections can directly impact yield and reliability [1]–[3].

Class-specific performance data (Table IV) highlights a significant 'generalization gap' between structural defects and contamination. While lUdc achieved an impressive mAP50-95 of 0.72, the Ptc class struggled with a precision-heavy but localization-poor profile (mAP50 = 0.591, mAP50-95 = 0.199). This disparity suggests that while the model is highly effective at identifying the presence of particles, it lacks the pixel-level boundary definition to precisely size them. Industrially, this indicates that while YOLOv11m is a robust 'trigger' for contamination alerts, further architectural refinement or higher magnification (e.g., 10×) may be required for high-precision metrology of particulate area.

Interestingly, increasing the training proportion to 80% (80/10/10 split) did not improve performance. Both architectures exhibited slightly reduced mAP50 and increased validation losses under this configuration. This indicates that insufficient validation data negatively impacted generalization monitoring, emphasizing the importance of maintaining adequate validation samples when working with limited datasets.

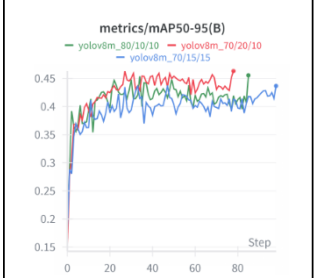
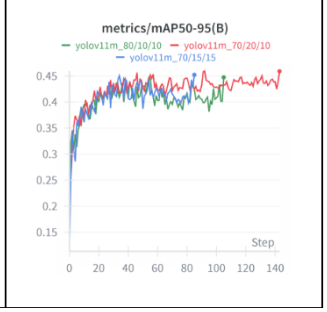
YOLO v8m with 3 Split Datasets Comparison	**YOLO v11m with 3 Split Datasets Comparison**
	

Fig. 4. Comparison of the mAP60-95(B) graph with different models and data splits.

C. Generalization Stability and Error Analysis

Across all configurations, the 70/20/10 split provided the best balance between training capacity and validation reliability, achieving the highest mAP50 and recall while demonstrating smoother validation convergence. Increasing the training proportion to 80% did not improve performance, indicating that reduced validation sampling negatively impacts model selection stability under limited data conditions.

Class-wise evaluation shows strong performance for lUdc (mAP50 = 0.956) and sUdc (mAP50 ≈ 0.91), while particle detection remains challenging (mAP50 ≈ 0.59). Qualitative inspection of YOLOv11m predictions under the 70/20/10 configuration reveals occasional confusion between background texture and particles, likely due to intra-class variability and limited dataset diversity. Particle detection remains challenging due to small size and variable morphology [1], [3]. Despite this limitation, recall for structural undercut defects remains high, which is critical for fabrication reliability.

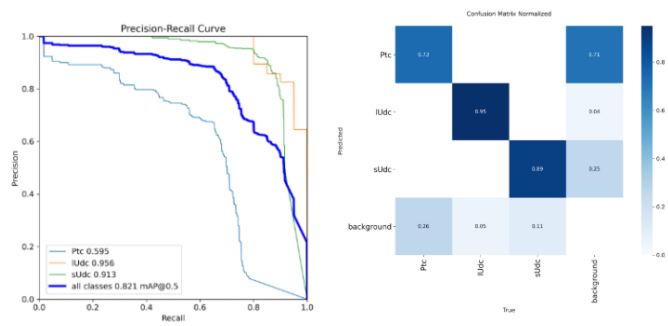


Fig. 5. PR Curve and confusion matrix of YOLO v11m 70% Train 20% Valid 10% Test.

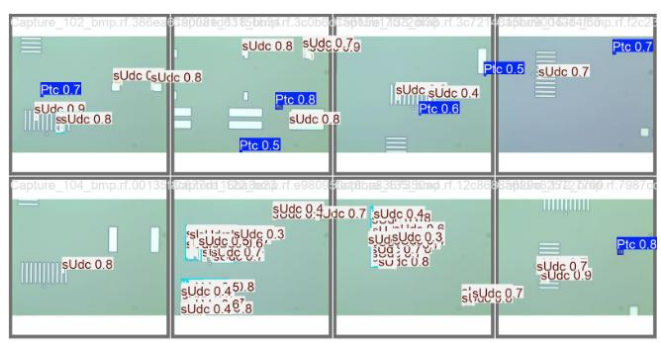


Fig 6. Prediction of YOLO v11m 70 20 10 data split.

V. Discussion

This study provides a clear articulation of the intellectual trade-off between training volume and validation reliability in small-scale industrial datasets. The results demonstrate that allocating a larger 20% validation set (60 original images) provides a more rigorous and 'trustworthy' supervisory signal than maximizing the training set to 80%. This higher validation threshold forces the model to prove its generalization across more unseen data during every epoch, significantly reducing

the risk of stochastic convergence artifacts that often plague small datasets.

Both YOLOv8m and YOLOv11m achieved stable convergence across all dataset partitions; however, clear differences emerged in generalization performance. The 70%/20%/10% split using YOLOv11m achieved the strongest results (mAP50 = 0.819, Recall = 0.810), indicating that a larger validation set improves model selection stability and reduces overfitting in limited-data conditions. Increasing the training proportion to 80% did not improve performance, suggesting that validation feedback is more critical than marginal gains in training volume when the dataset size is constrained.

Across most splits, YOLOv11m outperformed YOLOv8m in both mAP50 and recall, consistent with architectural refinements reported in improved YOLO variants that enhance feature fusion and small-object localization [7]. Higher recall is particularly important in semiconductor inspection, where missed defects directly affect manufacturing yield [1]–[3].

Although mAP50 performance was strong, the observed gap between mAP50 and mAP50–95 (≈0.36–0.38) indicates that high-IoU localization precision remains an area for improvement, as discussed in object detection evaluation literature [14]. Advanced localization loss formulations, such as Generalized Focal Loss, have been proposed to improve high-IoU bounding box regression accuracy [16], suggesting a potential direction for future refinement. Overall, the results confirm that under limited-data conditions, validation proportion plays a more critical role in generalization stability than maximizing training size.

VI. Conclusion

This study demonstrates that in the high-precision domain of semiconductor defect inspection, generalization stability is governed more by validation integrity than by training volume. Our findings challenge the conventional "more is better" approach to dataset partitioning, particularly when working with limited annotated samples. Experimental results confirm that a 70/20/10 split provides a significantly more reliable supervisory signal than an 80/10/10 configuration, with YOLOv11m achieving a peak mAP50 of 0.819 and a recall of 0.810. By maintaining a robust 20% validation buffer, we effectively mitigate the "noisy" convergence and stochastic artifacts that often plague small-scale industrial datasets, ensuring that early stopping is triggered by true model generalization rather than localized sampling bias.

To further bridge the gap between laboratory models and fab-floor requirements, future work will explore weighted loss formulations specifically designed to minimize critical false negatives in particulate contamination. Additionally, transitioning from standard horizontal detection toward oriented bounding boxes (OBB) will enable more granular directional etch analysis, providing engineers with diagnostic data on isotropic bias and tool-induced non-uniformity. Finally, multi-seed cross-validation will be standardized in future benchmarks to further quantify the statistical robustness of deep learning architectures in high-stakes semiconductor metrology.

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